

Discover once. Replay with confidence.

A working integration layer for business software whose only usable interface is its UI.

THE PROBLEM

A bank employee can complete a servicing task by clicking through screens. An agent needs the same ability when the vendor offers no business integration API.

01 Discover

AI observes the screen and proposes each next action.

02 Save

Verified success becomes a typed, reusable capability.

03 Replay

New inputs. Fixed steps.
No model in the loop.

The engine is the project. Harbor Ledger Bank is the synthetic target used to demonstrate it. You do not have to perform the clicks yourself before discovery.

6

banking workflows

141

passing local tests

0

model calls on replay

Start with the no-key replay on page 3. Then inspect a discovery, change the inputs, and try the same-session human takeover. All records are fictional; every flow stops at review.

Public repository and source

What each part does

One shared executor keeps policy, browser actions, result checks, and control ownership consistent.

Operator dashboard

Choose workflow and inputs. Inspect activity and results.

Banking website

The separate target UI, operated through Chromium.

Goal + masked screen > model proposal > checked UI action

Gemini receives the goal, typed contract, approved visible controls, recent context, and masked screenshot. It chooses one next action. The engine validates that proposal, binds the input values, and acts through Playwright. Then it observes again.

Discovery produces the artifact

A finish proposal is not proof. The engine reads the final UI and checks its values. Only successful, unassisted discovery saves a capability.

Replay consumes the artifact

Typed inputs replace parameter references. Fixed actions and declared recovery rules run without initializing a model client.

What developers still write: the adapter, allowed controls, goal/output contracts, state detectors, and error rules. The model discovers the order of actions; it does not build the whole integration from nothing.

Why AI here? It can discover an initial UI sequence that would otherwise need manual scripting. Repeated invocation then avoids repeated reasoning. The bank website itself works independently of AI.

Try a replay without a key

Install once, start both servers with one command, and use the dashboard for the demonstration.

```
git clone https://github.com/Maniss-ai/computer-use-capabilities.git
cd computer-use-capabilities
uv sync --locked --extra dev
export PLAYWRIGHT_BROWSERS_PATH="$PWD/.browser-cache"
uv run playwright install chromium
uv run capabilities web
```

Requires Python 3.12 and uv. On Linux, install browser system libraries with `uv run playwright install --with-deps chromium`.

1 Open Capability Studio

Visit <http://127.0.0.1:8766/>. The same command also starts the banking sandbox on port 8767.

2 Choose the bundled capability

Select **Sub-account review**, **Replay capability**, and **Prepared sub-account review**. This artifact came from a genuine recorded Gemini run.

3 Use a different member

Click **Member 10042**. Verify **10042 / Checking / Travel Fund**. Leave Test scenario on **Normal workflow**. Click **Run capability**.

4 Check the actual result

Expect **Success**, **Model calls = 0**, and **Review sub-account**. Result must show all three input values and **review ready = true**. Activity lists checkpoints; Capability shows the JSON.

Success stops at review. No account is opened and no banking record changes. The live view shows the engine-owned browser, not a pre-rendered animation.

Let AI find the path

No manual recording or demonstration is needed. Supply the goal inputs and let the agent operate the UI.

1 Configure discovery access

Add `GEMINI_API_KEY=...` to the ignored local `.env` file and restart the dashboard. Keep the key out of Git. Replay and tests need no key.

2 Start one genuine discovery

Choose **Sub-account review > Discover with AI > Gemini**. Click **Member 10023: 10023 / Savings / Emergency Fund**. Keep Normal workflow, then click **Discover and save**.

3 Verify before reusing

Wait for Success, a matching review, nonzero model calls, and a new saved capability. If the run stops, inspect Result and Activity; a failed discovery must not publish an artifact.

4 Replay that same capability

Click **Use this capability with new inputs**. Keep the new capability selected. Click **Member 10042**, then **Run capability**. Confirm new values and zero model calls.

RECORDED EXAMPLE / SUB-ACCOUNT

Discovery: 7 calls / 86.2s

Replay: 0 calls / 5.3s

One observed pair after the timeout fix; timings vary. This successful live discovery did not need a retry.

If Gemini is slow: transient request failures have bounded retries of the undecided step; completed UI actions are not repeated. Quota and authentication errors stop. There is no paid-provider fallback.

The workflow library counts saved artifacts, not trained models. A fresh checkout bundles one sub-account capability. Each workspace adds its own discoveries.

Six realistic review workflows

A = discover with Member 10023. B = replay the saved capability with Member 10042.

01 Sub-account review

Prepare an additional Savings or Checking account request.

Member ID	10023	10042
Product	Savings	Checking
Account nickname	Emergency Fund	Travel Fund

02 Card replacement

Find a member-owned card and prepare replacement preferences.

Member ID	10023	10042
Card reference	CARD-10023	CARD-10042
Replacement reason	Damaged	Lost
Replacement delivery	Branch pickup	Registered address

03 Transaction dispute

Find a posted payment and prepare an investigation intake.

Member ID	10023	10042
Transaction reference	TXN-10023	TXN-10042
Dispute reason	Duplicate charge	Service not received
Contact channel	Secure message	Phone

Expected for every normal case: the matching review screen, all requested values, review ready = true, and no final submission. Replay always has zero model calls.

Member 10077 (Avery Patel) is also available. Record references must match the member: CARD-, TXN-, ACC-, or FEE- followed by the member ID.

Profile, documents, and fees

A = discover with Member 10023. B = replay the saved capability with Member 10042.

04 Address change

Prepare a mailing address change and a verification route.

Member ID	10023	10042
Street address	45 Market Street	280 Pine Avenue
City	San Francisco	Austin
State code	CA	TX
Postal code	94105	78701
Verification method	Branch verification	Callback verification

05 Statement request

Prepare a statement period, document format, and delivery request.

Member ID	10023	10042
Account reference	ACC-10023	ACC-10042
Statement period	September 2026	August 2026
Document format	PDF	Paper copy
Document delivery	Secure inbox	Branch collection

06 Fee adjustment

Find an assessed fee and prepare a waiver-review request.

Member ID	10023	10042
Fee reference	FEE-10023	FEE-10042
Adjustment reason	First occurrence	Service issue
Requested resolution	Full waiver	Partial waiver
Supporting note	First fee review requested by member	Service issue reported to branch

Handle exceptions. Keep control.

Use the bundled sub-account capability with 10042 / Checking / Travel Fund. Change only the item shown.

Missing member

Member ID = 99999

Outcome: member_not_found. A valid business answer; no review.

Known notice

Scenario = Known notice

Automatic recovery, then Success. Zero model calls.

Permission denied

Scenario = Permission denied

Stopped with a permission failure. No attempt to bypass access.

Session expiry

Scenario = Session expired

Needs you. Claim, restore in the live preview, and return control.

SAME BROWSER. ONE OWNER AT A TIME.

Click Claim session. Click Restore session inside the live banking preview. Click Return to automation. The engine rechecks the screen and resumes. A control epoch rejects stale requests and prevents concurrent human/agent actions.

Do not use Open manual sandbox for takeover. That link opens an independent session. Manual actions there do not train discovery or repair the engine-owned browser.

What is proven, and where to look

The evidence is reviewable. Fixture tests and genuine model runs are labeled separately.

141 local tests passed

Lint, format, strict typing, real-browser tests, and submission evidence checks pass. The live gate requires genuine discovery and a successful zero-model replay with the same artifact hash.

Core loop and artifact

evidence/live/ + schemas/capability.schema.json

Error handling and handoff

evidence/offline/ + tests/test_dashboard.py

Expanded genuine runs

evidence/expanded/ (card replacement and dispute)

Timeout fix and matching replay

evidence/timeout-recovery/

Reasoning and deliberate cuts

REPORT.md (seven required assignment headings)

Coverage: all six goals pass browser tests with scripted planners. Sub-account, card replacement, and dispute also have successful genuine Gemini discovery/replay evidence. Address, statement, and fee do not yet have successful live-model evidence. Failed runs are retained.

Scope: synthetic data; one active run; local console; files for artifacts/evidence; in-memory target records. Desktop drivers, production authentication, and multi-tenant runtime are design extensions. This is not a deployed production banking service.

Sharing: GitHub hosts the repository, guide, and evidence. GitHub Pages cannot run Python/Chromium. A full online demo needs a runtime such as Codespaces or a separate authenticated server host.

[Repository](#)

[Tests and CI](#)

[Full guide + test inputs](#)